INSTRUCTION

Follow the directions on the left side of your screen, and respond to the prompts in this sheet. Enter your responses in the white cells and replace "YOUR WORK HERE" with your response to each question.

STEP 1 - Explore the Data

Question 1	**How many rows and columns of data are there?** *NOTE:* Exclude *Number of Reviews Per Day* from your count.	
	Rows	5551
	Columns	16
Question 2	**Categorize each column from the *DataExport_2020* sheet into numerical or categorical data types.** *NOTE:* Exclude *Average Number of Reviews Per Day* from your categorization.	
	Numerical	Bathrooms, Bedrooms, Beds, Price, Number of Reviews, Days Listed, Review Score Location, Review Score Value
	Categorical	Host ID, Cancellation Policy, City, State, Room Type, Bed Type, Amenities
Question 3	**How many quantitative columns are there? How many qualitative columns are there?**	
	Quantitative	8
	Qualitative	6
Question 4	**In the *Number of Reviews Per Day* column, you should see some cells that contain " #DIV/0!" Click on these cells. You should be prompted with an error message. Although you have not explicitly learned the error message in the course, you should be able to determine why it is happening by examining the values in the *Days Listed* column.**	
	What is the error message?	
	"Function DIVIDE parameter 2 cannot be 0."	
	What is the error message telling us? Why could it be happening? What values in the *Days Listed* or *Number of Reviews Per Day* columns could be affecting this? ***HINT:*** **"parameter 2" = denominator**	
	Days Listed values cannot be 0. This means, when you divide Number of Reviews by Days Listed, it cannot be divided by 0.	

STEP 2 - Summarize Data Using A Pivot Table

Question 1	**Based on your *Summary By City* pivot table, which city has the highest average prices for listings? lowest average prices for listings?**	
	Highest	Austin
	Lowest	Oakland
Question 2	**Let's look at the city Austin. How many total listings are there in Austin? How many unique hosts are there in Austin?** **Is this the city with the most listings? If, not, what city is that?**	
	# Listings	868
	# Hosts	777
	city with Max # Listings	San Francisco

Question 3	**What city has the highest percentage of listings with repeat hosts? lowest percentage of listings with repeat hosts?**	
	Highest	Boston
	Lowest	Seattle

STEP 3 -Clean Inconsistent, Invalid, Duplicate, and Missing Data

Question 1

How many empty cells are there in each column?
HINT: Use =COUNTBLANK()

Bathrooms	Bedrooms	Beds	Amenities	Review Scores Location	Review Scores Value
7	4	5	2	1	1

STEP 4 - Clean Text Data I (Search and Count)

Question 1

Between *Internet*, *Kitchen*, *24-Hour Check-In*, *Free Parking*, *Private Entrance*, and *Smoking Allowed*, which two were most common? Two least common?
NOTE: Use formulas to calculate this information in the spaces below.

Internet	Kitchen	24-Hour Check-In	Free Parking	Private Entrance	Smoking Allowed	Other
5351	4995	2472	2922	493	219	50

2 Most Common	Internet & Kitchen
2 Least Common	Smoking Allowed & Other

STEP 5 - Clean Text Data II (Split Text to Columns)

Question 1

What is the maximum number of amenities advertised by a listing? minimum number of amenities?
NOTE: Use functions to answer Question 1.

MAX	57
MIN	0

Question 2

How many unique amenities are there listed in Listings123?
HINT: Use your *Scratch Work* sheet to answer this question.

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STEP 6 - Merge Tables

Question 1

How many rows and columns of data are there in your *Merged City Data* table?

Rows	10
Columns	14

Question 2

What city has the highest number of full-service restaurants? lowest number of full-service restaurants?
NOTE: Use the *Merged City Data* sheet to answer this question.

Highest	Chicago
Lowest	New Orleans

STEP 7 - Summarize Data Wrangling

In the space below, document the data wrangling steps you have completed in your analysis so far.
TIP: To create a new line press *CMD + Enter (Mac)* OR *CTRL + Enter (Windows).*

Question 1	1. Filled in all missing values by dividing Number of Reviews by Days Listed in column P of DataExport_2020. 2. Calculated column L by using the formula =(B2-C2)/B2*100 for Summary By City. 3. Replaced invalid and inconsistent values: '-1', 'none', '0.1', '-1000', and '321' with (blanks) in columns E, F, and D of Copy of DataExport_2020. 4. Replaced missing values: 'none' with 1 in columns E, F, and D of Copy of DataExport_2020 (For only Room Type: Shared Room and Private Room). 5. Deleted entire rows with empty values: 'none' in column I, V, and W of Copy of DataExport_2020. 6. Removed duplicate rows in Copy of DataExport_2020. 7. Fixed #DIV/0! errors by wrapping the calculated field with an =IF(function) in column X of Copy of DataExport_2020. 8. Replaced missing values: 'translation missing:en.hosting_amenity_49' and 'translation missing:en.hosting_amenity_50' with (blanks) in column I of Copy of Data Export_2020. 9. Filled in all missing values with an =ISNUMBER(SEARCH(function)) in columns L, M, N, O, and P in Copy of DataExport_2020. 10. Split text into columns by referencing column I of Copy of DataExport_2020 for Scratch Work. 11. Filled in all missing values with an =COUNTA(function) for column Q of Copy of DataExport_2020. 12. Pasted Restaurants and Hotels reference into Restaurants Hotels. 13. Filled in all missing values in column C by referencing the information from city-data.com of Restaurant Hotels 14. Merged Summary By City with Restaurants Hotels. 15. Filled in all missing values with an =VLOOKUP(function) for columns L and M of Merged City Data by referencing from Restaurant Hotels. 16. Added a New Orleans, Lousiana data row by referencing the information from city-data.com of Restaurant Hotels. 17. Fixed #N/A error by referencing Restaurant Hotels of for columns M and N of Merged City Data.